Lab 9 — Provisioning an Azure Databricks Environment Manually

Description

You will build a VNet-injected Azure Databricks workspace from nothing, using the Azure portal only. Allow 45 minutes; workspace deployment alone takes several minutes. You need Contributor on the sandbox subscription and the Network Contributor role on the virtual network, or a custom role granting subnet join and write permissions.

Learning Objectives

- Build the network prerequisites for a VNet-injected workspace in the correct order.
- Deploy a classic workspace into your own virtual network.
- Verify delegation, network security group rules and workspace health after deployment.
- Describe what the portal configured on your behalf, and what it did not.

Step by Step Guide

1. Create a resource group. In the Azure portal, create a resource group named for yourself, for example rg-adb-<your-name>, in the region your instructor specifies. Everything you build in this lab goes here.
2. Create a network security group. Create an empty NSG in the same resource group and region. Add no rules — Databricks provisions the ones it needs.
3. Create the virtual network. Create a VNet in the same region and subscription as the workspace, with an address space between /16 and /24.
4. Add the two subnets. Within the VNet, create a host subnet and a container subnet, each /26 or larger. Attach the NSG you created to both. Leave subnet delegation unset — the workspace deployment applies it. If the portal requires a delegation, set it to Microsoft.Databricks/workspaces.
5. Add outbound connectivity. Create a NAT gateway in the same region and associate it with both subnets. New Azure virtual networks no longer provide default outbound internet access, so without this your clusters cannot reach the internet.
6. Start the workspace deployment. Select Create a resource, then Analytics, then Azure Databricks. On the Basics tab set the workspace name, your subscription, the resource group from step 1, the same region, Pricing Tier of Premium, and Workspace type of Hybrid — Hybrid creates a classic workspace, which VNet injection requires.
7. Configure networking. On the Networking tab, choose to deploy the workspace in your own virtual network. Select your VNet, then enter the exact names and IP ranges of your two subnets. Subnet CIDR ranges cannot be changed after deployment.
8. Deploy. Select Review + create, then Create. Watch the status move from Provisioning to Running. This takes several minutes — continue to the next step only once it reports Running.

9. Verify the workspace. Open the workspace resource. Confirm it is healthy rather than failed, that both the resource group and the managed resource group are listed, and that virtual network peering is disabled — which is expected for VNet injection.
10. Verify the subnets. Return to the VNet and open Subnets. Confirm both subnets now show a network security group attached and a delegation to Microsoft.Databricks/workspaces that you did not set yourself.
11. Verify the NSG rules. Open the NSG. It is no longer empty — review the inbound and outbound rules Azure Databricks added, and note that you must not modify or delete them.
12. Prove it works. Launch the workspace, create a small compute resource with a short auto termination period, and wait for it to start. Then open the managed resource group and confirm virtual machines, disks and network interfaces have appeared, with interfaces in both of your subnets.
13. Clean up. Terminate the compute resource. Delete the workspace — this also deletes the managed resource group. Then delete your resource group.

Outcome

You have deployed a working VNet-injected workspace end to end and verified it against the documented checks. You can state which parts of the configuration you supplied and which parts Databricks applied automatically through subnet delegation.